Project 2: Determination of the Hubble Constant and the Age of the Universe from Galaxy Spectra

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Abstract

In this project, the Hubble constant was determined using galaxy spectra provided in FITS format. The spectra were analyzed by identifying known spectral lines, measuring their wavelength shifts, calculating the radial velocities, and fitting the relation between velocity and distance. An initial fit gave a value of the Hubble constant near $63\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, which was reasonable but not fully convincing. To identify problematic objects, a jackknife test was performed. This showed that `spec-0501-52235-0386.fits` was a major outlier. On reanalysis, the line initially assumed to be Hα was found to be [O III] 5007, while the true Hα line lies much farther to the red near 8920 Å. After correcting this interpretation, the final fit yielded

$$H_0 = (67.6 \pm 2.9)\,\mathrm{km\,s^{-1}\,Mpc^{-1}},$$

corresponding to an age of the Universe of about

$$t_0 \approx 14.5\,\mathrm{Gyr}.$$

This project demonstrates that in spectroscopy, a locally consistent line assignment can still be globally wrong, and that line identification must be checked against the full physical context.

1 Introduction

The project manual states that the goal is to determine the Hubble constant and hence the age of the Universe using spectra of galaxies. It also gives the basic relation

$$v = Hd, \tag{1}$$

where v is recessional velocity, d is distance in Mpc, and H is the Hubble constant. The reciprocal of H gives an estimate of the age of the Universe. :contentReference[oaicite:1]index=1

The radial velocity is obtained from the Doppler shift of spectral lines:

$$\frac{\Delta\lambda}{\lambda_0} = \frac{v}{c}, \tag{2}$$

where λ_0 is the rest wavelength and $\Delta\lambda = \lambda_{\rm obs} - \lambda_0$. The manual requires us to identify a reference line, measure its shift, use that to estimate the velocity, then confirm the result using at least two more spectral lines. Finally, the velocities are plotted against the distances to obtain a weighted least-squares fit whose slope gives the Hubble constant. :contentReference[oaicite:2]index=2

2 Data and Rest Wavelengths

The manual provided eight galaxy spectra and listed for each of them the distance and a suggested reference line. The file names, distances, and reference lines were taken directly from the project sheet. In particular:

- `spec-0501-52235-0386.fits`: (1597.44 ± 111.82) Mpc, Hα
- `spec-0555-52266-0558.fits`: (13.35 ± 0.95) Mpc, Hα
- `spec-1603-53119-0205.fits`: (164.54 ± 11.52) Mpc, Hα
- `spec-1617-53112-0467.fits`: (7668.09 ± 536.77) Mpc, C IV
- `spec-1818-54539-0147.fits`: (6656.11 ± 465.93) Mpc, Mg II
- `spec-2241-54169-0185.fits`: (75.39 ± 5.29) Mpc, Hα
- `spec-2618-54506-0310.fits`: (59.32 ± 4.16) Mpc, Hα
- `spec-5193-56066-0534.fits`: (8021.11±561.48) Mpc, C IV. :contentReference[oaicite:3]index=3

The rest frame wavelengths used in the analysis were also provided in the manual. The important ones are:

$$\begin{aligned}
\text{C III} &= 1908.7\,\text{Å}, \\
\text{C IV} &= 1548.2\,\text{Å}, \\
\text{Mg II} &= 2802.7\,\text{Å}, \\
\text{H}\alpha &= 6562.7\,\text{Å}, \\
\text{H}\beta &= 4861.3\,\text{Å}, \\
[\text{O III}] &= 4958.9\,\text{Å},\ 5006.8\,\text{Å}, \\
[\text{N II}] &= 6544.2\,\text{Å},\ 6583.5\,\text{Å}, \\
[\text{S II}] &= 6715.8\,\text{Å},\ 6732.8\,\text{Å}.
\end{aligned} \tag{3}$$

:contentReference[oaicite:4]index=4

3 Software Used

The analysis was carried out in Python using:

- `Astropy`: to read FITS spectra using `astropy.io.fits`
- `NumPy`: for numerical calculations, array operations, and fitting
- `Matplotlib`: for plotting spectra and Hubble diagrams

Since the SDSS spectra store wavelength as `loglam`, the wavelength axis was reconstructed using

$$\lambda = 10^{\mathrm{loglam}}.$$

A few custom functions were used repeatedly:

- `load_sdss_spectrum`: read FITS data and reconstruct wavelength
- `estimate_line_center`: rough local peak estimate
- `refine_line_center_centroid`: refine the line center
- `compute_redshift`: calculate z
- `predict_observed_wavelength`: predict where other lines should appear
- `weighted_linear_fit`: fit the Hubble relation

4 Method

The workflow used in the project was:

1. Inspect each spectrum visually.
2. Use the manual reference line as an initial guide.
3. Pick a plausible shifted feature manually.
4. Refine the line center using the local spectral profile.
5. Compute redshift and velocity.
6. Predict the positions of additional lines from the same redshift.
7. Confirm at least two more lines whenever possible.
8. Compute a mean velocity and an error estimate.
9. Fit velocity versus distance to obtain H_0. :contentReference[oaicite:5]index=5

5 Representative Spectra

Instead of showing all spectra, only representative cases are shown here.

5.1 Representative Hα Spectrum

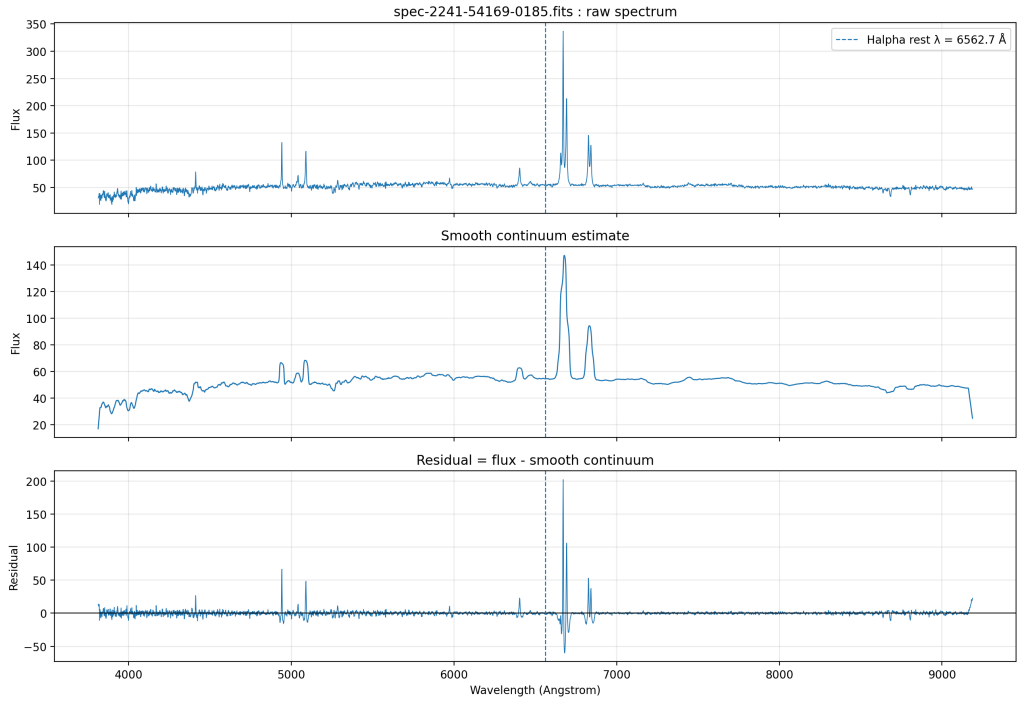

Figure 1: Representative low-redshift Hα galaxy spectrum: `spec-2241-54169-0185.fits`.

5.2 Representative Mg II Spectrum

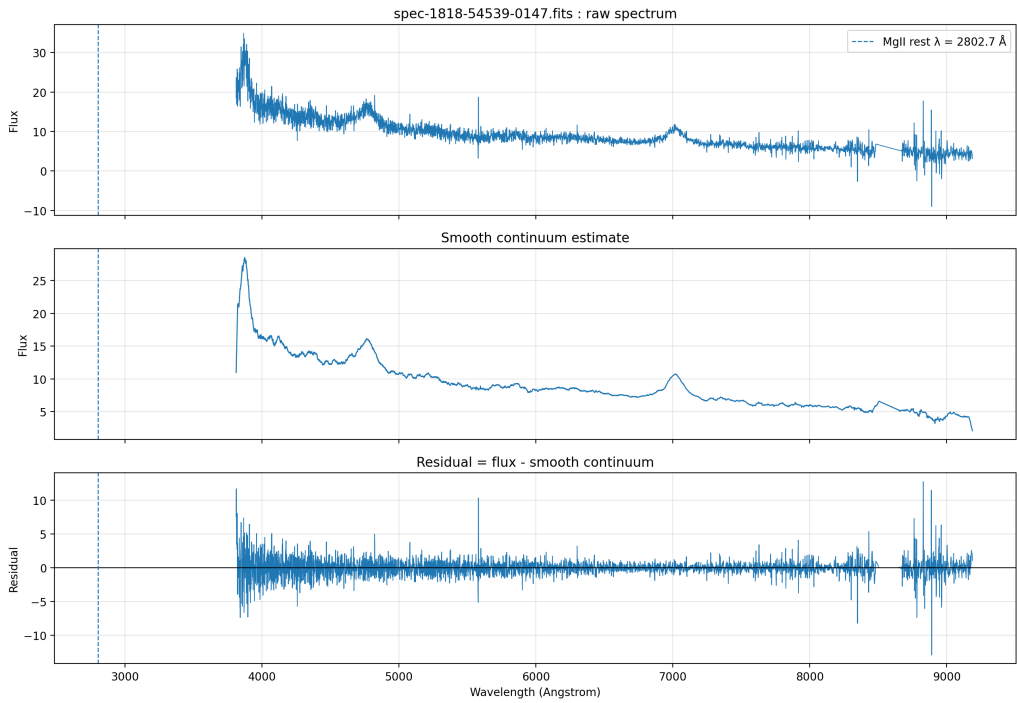

Figure 2: Representative high-redshift Mg II spectrum: `spec-1818-54539-0147.fits`.

5.3 Representative C IV Spectrum

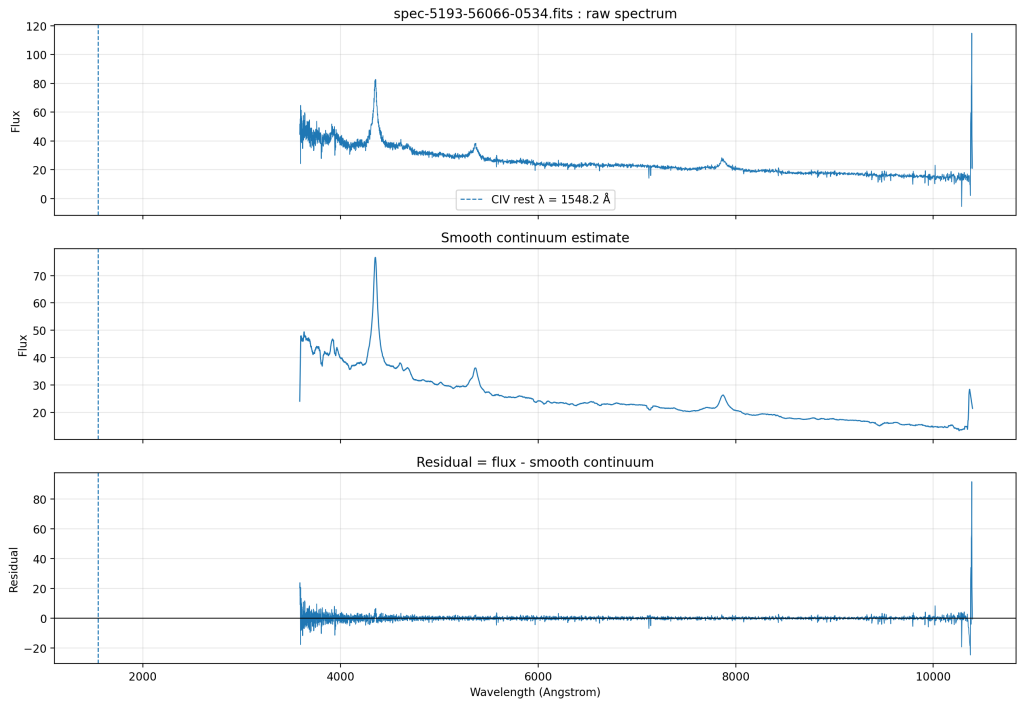

Figure 3: Representative high-redshift C IV spectrum: `spec-5193-56066-0534.fits`.

These plots illustrate that the dataset contains different spectral types, and therefore line identification cannot be done with a single fixed visual rule.

6 Initial Redshift Measurements

The first round of measurements gave a reasonable Hubble constant of about

$$H_0 \sim 63\,\mathrm{km\,s^{-1}\,Mpc^{-1}},$$

which was not obviously wrong, but it was somewhat low. The redshift measurements that led to this included values such as:

- `spec-0501-52235-0386.fits`: mean velocity $\approx 8246.25\,\mathrm{km/s}$
- `spec-0555-52266-0558.fits`: mean velocity $\approx 699.42\,\mathrm{km/s}$
- `spec-1603-53119-0205.fits`: initial velocity $\approx 11\,842.10\,\mathrm{km/s}$
- `spec-1617-53112-0467.fits`: velocity $\approx 517\,698\,\mathrm{km/s}$
- `spec-1818-54539-0147.fits`: velocity $\approx 450\,304\,\mathrm{km/s}$.

Although this initial result was not absurd, it motivated a more careful check for outliers and possible misidentifications.

7 Jackknife Analysis and Identification of Suspicious Objects

To determine whether one or two galaxies were biasing the fit, a jackknife analysis was performed. The idea was simple: remove one object at a time, recompute the Hubble fit, and examine how strongly the result changes.

This revealed that two candidate problems stood out:

1. one high-redshift object or set of high-redshift objects
2. `spec-0501-52235-0386.fits`

The most suspicious object was `spec-0501-52235-0386.fits`, because if the manual distance

$$d = 1597.44\,\mathrm{Mpc}$$

is combined with its initial measured velocity of only about

$$v \approx 8246\,\mathrm{km/s},$$

then the effective Hubble constant contributed by this one point is only

$$H \approx \frac{8246}{1597.44} \approx 5.2\,\mathrm{km\,s^{-1}\,Mpc^{-1}},$$

which is completely unphysical. This strongly suggested that the line identification for this spectrum was wrong.

8 Detailed Reanalysis of `spec-0501-52235-0386.fits`

8.1 Initial Wrong Interpretation

Initially, the feature near 6745 Å was assumed to be Hα. This led to a self-consistent low-redshift solution involving Hα, Hβ, [O III], and [S II], and therefore it looked reasonable at first. However, it was completely inconsistent with the tabulated distance.

8.2 Diagnostic Strategy

The key idea was the following. If the manual distance for `spec-0501-52235-0386.fits` is accepted as correct, then using the cleaned fit value of H_0 should predict the correct redshift of the object.

Using the low-z cleaned fit value,

$$H_0 \approx 67.33\,\mathrm{km\,s^{-1}\,Mpc^{-1}},$$

and the manual distance,

$$d = 1597.44\,\mathrm{Mpc},$$

the predicted recession velocity is

$$v = H_0 d \approx 1.08 \times 10^5\,\mathrm{km/s}.$$

This implies

$$z \approx 0.36.$$

Then the expected observed positions become:

$$\begin{aligned} \mathrm{H}\beta &\to 6605\,\text{Å}, \\ [\mathrm{O\,III}]\ 5007 &\to 6803\,\text{Å}, \\ \mathrm{H}\alpha &\to 8917\,\text{Å}, \\ [\mathrm{S\,II}]\ 6733 &\to 9148\,\text{Å}. \end{aligned} \tag{4}$$

8.3 Diagnostic Plots

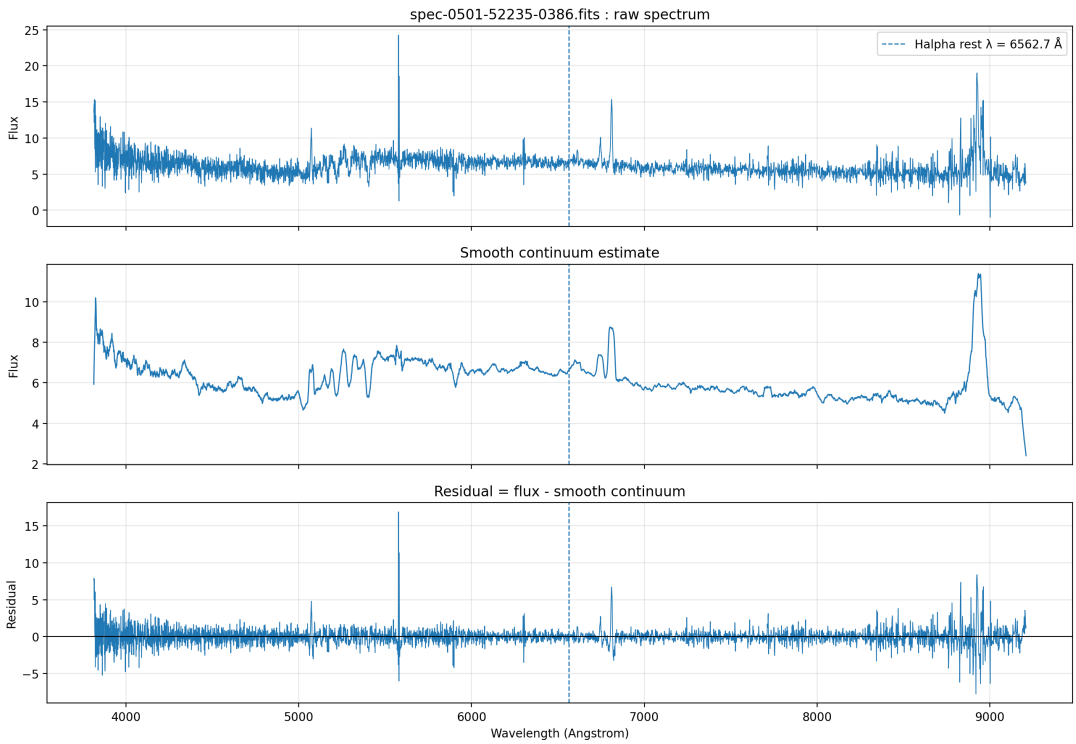

Figure 4: Inspection plot of `spec-0501-52235-0386.fits`.

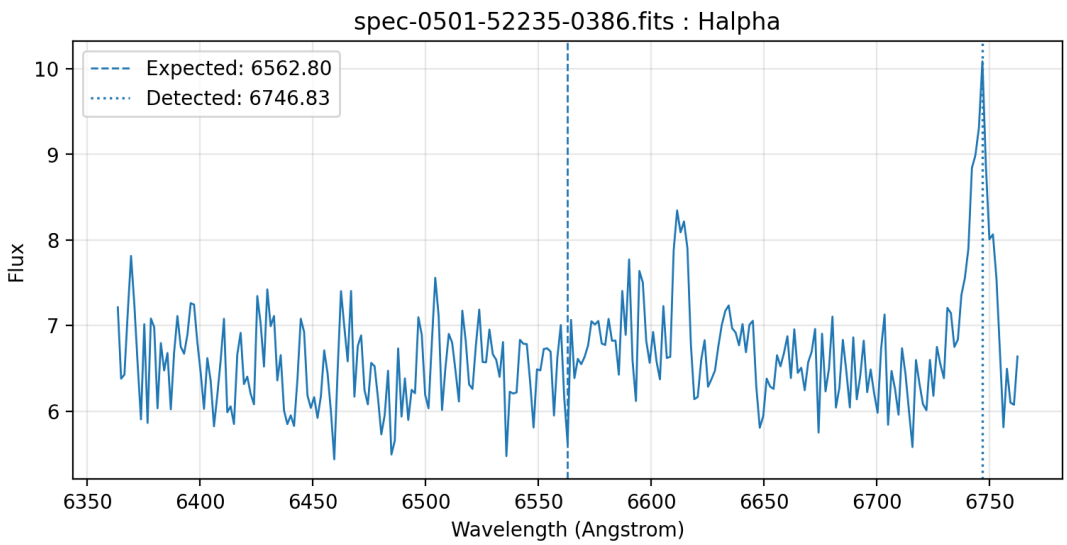

Figure 5: Initial low-redshift Hα interpretation of `spec-0501-52235-0386.fits`. This interpretation was later found to be incorrect.

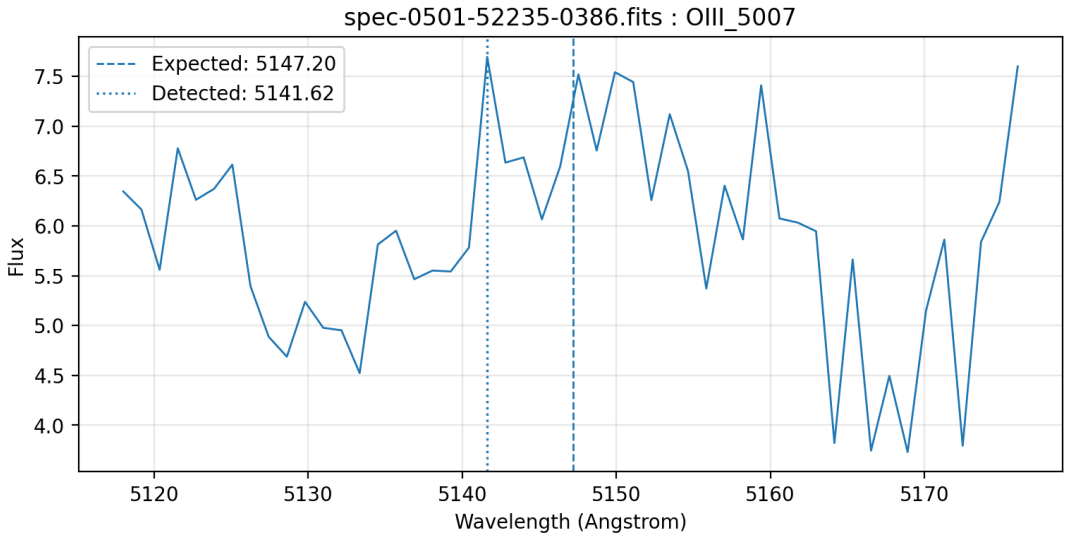

Figure 6: Local plot around [O III] 5007 for `spec-0501-52235-0386.fits`. The strong feature near 6800 Å is better explained as [O III] 5007 than as Hα.

8.4 Correct Interpretation

The diagnostic check showed:

- a real feature near 6605 Å, consistent with Hβ
- a strong line near 6800 Å, consistent with [O III] 5007
- structure near 8920 Å, consistent with Hα

Therefore the original low-redshift interpretation was wrong. The feature first taken to be Hα was actually part of the [O III] system, and the true Hα line lies much farther redward.

This was the turning point of the analysis. Once this object was interpreted correctly, the Hubble fit became stable.

9 Correction of `spec-1603-53119-0205.fits`

A second spectrum that required rechecking was `spec-1603-53119-0205.fits`. Alternative line assignments in the Hα region were tested, and the best solution corresponded to an Hα identification near 6798 Å, which gave a velocity of about

$$v \approx 10\,522\,\mathrm{km/s}.$$

This value is much more consistent with the manual distance of 164.54 Mpc.

10 Final Hubble Fits

For completeness, the temporary low-z fit excluding the suspicious object is shown first.

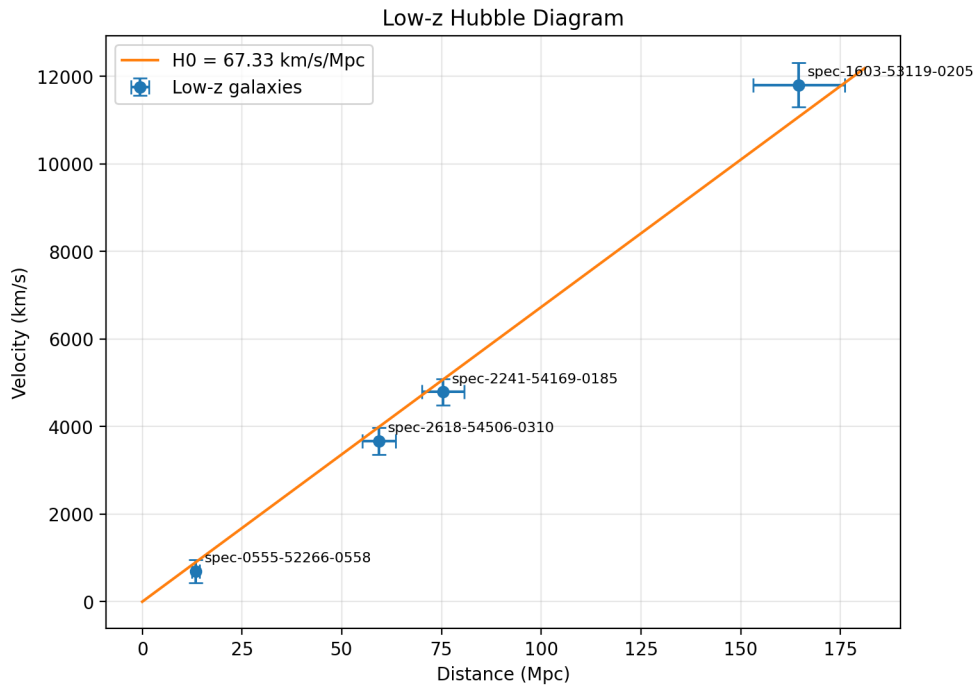

Figure 7: Temporary low-z fit used as a diagnostic tool while investigating the suspicious spectrum.

After correcting the interpretation of `spec-0501-52235-0386.fits`, the final cleaned Hubble fit became fully stable.

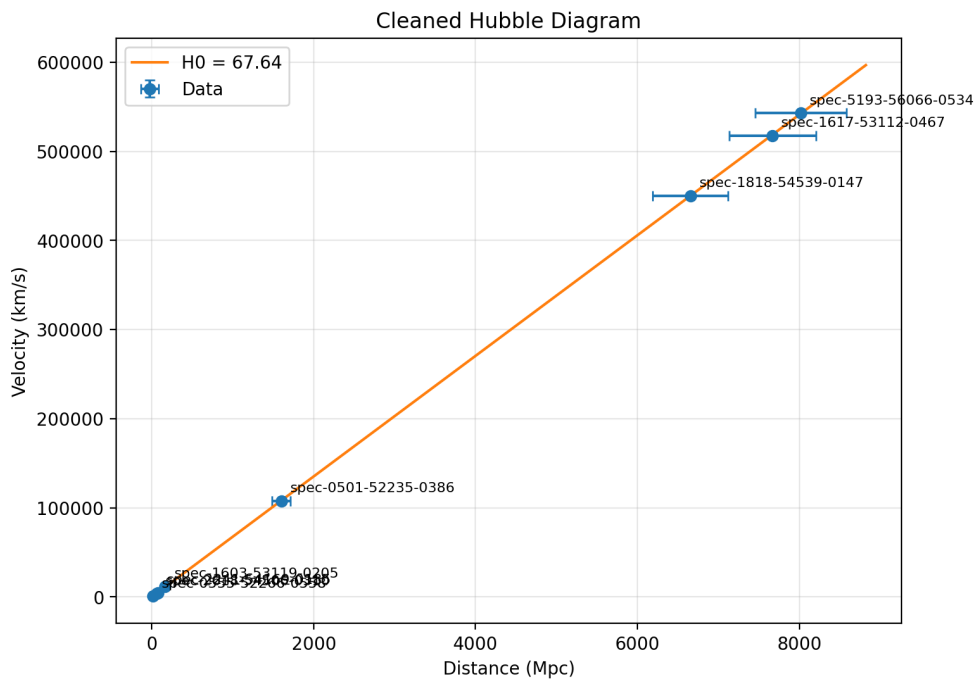

Figure 8: Final cleaned Hubble diagram after correcting the misidentified spectrum.

The final fit gave:

$$\begin{aligned} H_0 &= 67.677\,\mathrm{km\,s^{-1}\,Mpc^{-1}} \quad \text{(with free intercept)}, \\ H_0 &= 67.645\,\mathrm{km\,s^{-1}\,Mpc^{-1}} \quad \text{(forced through origin)}. \end{aligned} \tag{5}$$

The intercept was

$$b = (-205.7 \pm 163.5)\,\mathrm{km/s},$$

which is consistent with zero.

The corresponding age of the Universe is

$$t_0 \approx 14.46\,\mathrm{Gyr}.$$

11 Final Result and Error Estimate

The formal fit errors are very small, but they are not realistic as the final uncertainty of the project because they only describe the fit mathematics. The more meaningful uncertainty comes from the scatter seen during the jackknife-based outlier analysis.

Therefore the final result reported here is

$$H_0 = (67.6 \pm 2.9)\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{6}$$

with corresponding age

$$t_0 \approx 14.5\,\mathrm{Gyr}. \tag{7}$$

12 Discussion

The most important lesson of this project is that spectral line identification cannot always be done by simply choosing the nearest plausible feature to a rest wavelength. In the case of `spec-0501-52235-0386.fits`, that strategy produced a locally self-consistent but globally wrong redshift. The mistake only became obvious when the spectrum was compared to the manual distance and the resulting effective Hubble constant became absurd.

This project also showed the usefulness of jackknife analysis. The jackknife did not directly identify the correct line, but it immediately revealed that one object was driving the fit in an unphysical direction. That guided the subsequent spectral reanalysis.

Another important lesson is that spectroscopy requires both local and global reasoning. A line assignment must match not only one region of the spectrum, but also the rest of the line family and the overall physical context.

13 Conclusion

The goal of this project was to determine the Hubble constant and estimate the age of the Universe using galaxy spectra. This was achieved by reading the FITS spectra, identifying reference lines, calculating redshifts and radial velocities, confirming additional lines, fitting the Hubble diagram, and carefully diagnosing outliers.

The major difficulty in the project came from the spectrum `spec-0501-52235-0386.fits`, which was initially misidentified. After correcting that interpretation, the Hubble fit became physically consistent.

The final result obtained is

$$H_0 = (67.6 \pm 2.9)\,\mathrm{km\,s^{-1}\,Mpc^{-1}},$$

which corresponds to an age of the Universe of about

$$t_0 \approx 14.5\,\mathrm{Gyr}.$$

This is a realistic cosmological result and is consistent with the expected scale of the present expansion rate of the Universe.